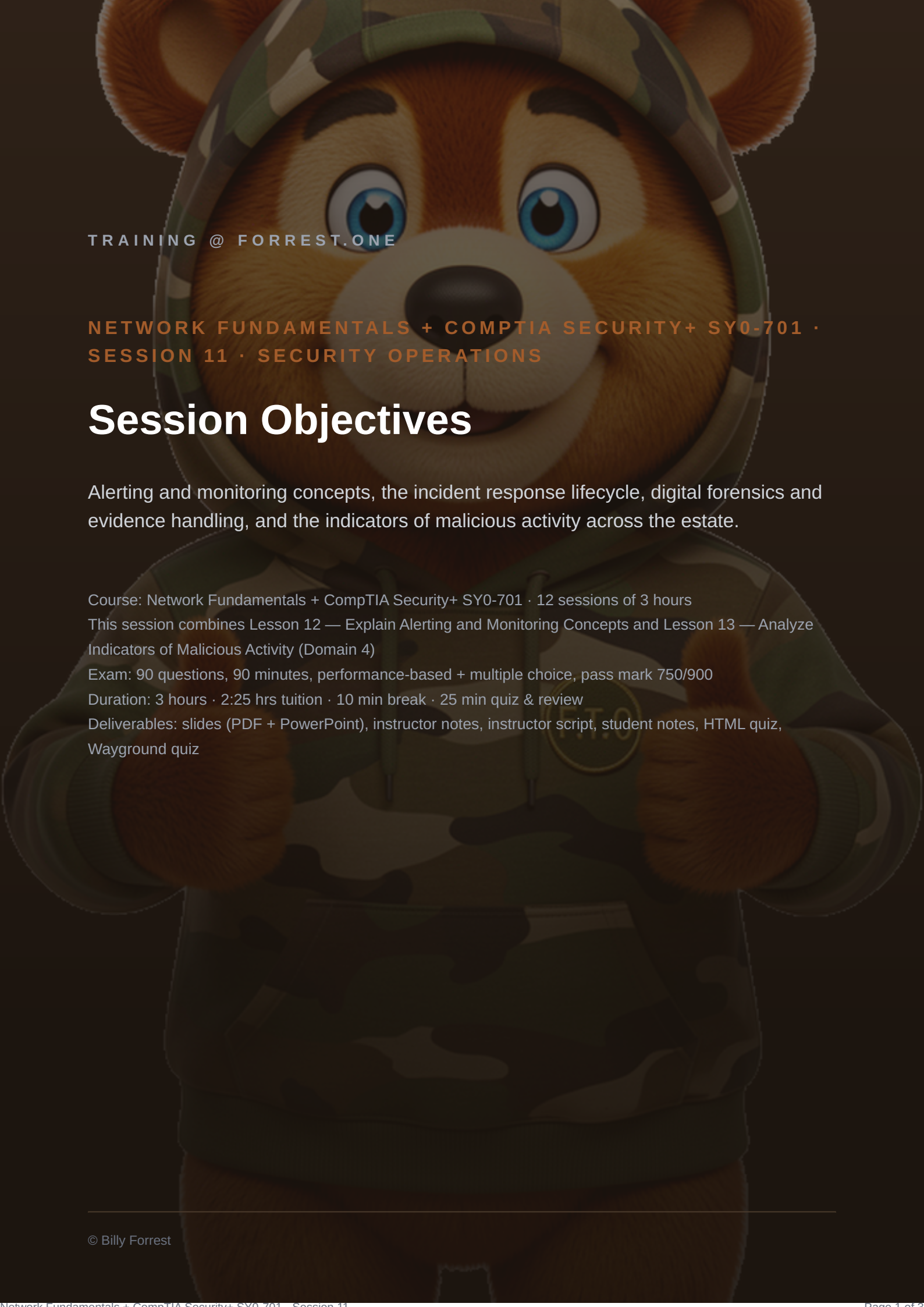




TRAINING @ FORREST.ONE

NETWORK FUNDAMENTALS + COMPTIA SECURITY+ SY0-701 ·
SESSION 11 · SECURITY OPERATIONS

Session Objectives

Alerting and monitoring concepts, the incident response lifecycle, digital forensics and evidence handling, and the indicators of malicious activity across the estate.

Course: Network Fundamentals + CompTIA Security+ SY0-701 · 12 sessions of 3 hours

This session combines Lesson 12 — Explain Alerting and Monitoring Concepts and Lesson 13 — Analyze Indicators of Malicious Activity (Domain 4)

Exam: 90 questions, 90 minutes, performance-based + multiple choice, pass mark 750/900

Duration: 3 hours · 2:25 hrs tuition · 10 min break · 25 min quiz & review

Deliverables: slides (PDF + PowerPoint), instructor notes, instructor script, student notes, HTML quiz, Wayground quiz

Session Objectives — Security Operations: Monitoring, Incident Response & Forensics

This session covers what happens when an incident occurs, combining Lessons 12 and 13 of the Cyber Academy programme and mapping to Domain 4 (Security Operations, 28% of the exam, the largest domain). Monitoring, response and forensics are examined heavily and directly. The session runs in six chapters: monitoring and network security devices, endpoint and data controls, identity and automation, incident response, threat hunting and forensic evidence, then logs and consolidation.

Learning outcomes

By the end of the 3-hour session, candidates will be able to:

- Compare firewall types: packet filtering, stateful inspection and the next-generation firewall (NGFW).
- Explain firewall rule creation, change control, regular review and the use of segmentation, including the demilitarized zone (DMZ).
- Describe the web application firewall (WAF) and the web attacks it blocks, including Structured Query Language (SQL) injection.
- Distinguish intrusion detection from intrusion prevention, and network-based from host-based deployment.
- Compare signature-based and anomaly-based detection, and explain tuning.
- Read a host log such as the xrdp Remote Desktop Protocol (RDP) session log for signs of brute force.
- Explain Domain Name System (DNS) filtering, data loss prevention (DLP) and network access control (NAC) as controls at different layers.
- Describe endpoint detection and response (EDR) and extended detection and response (XDR), and the alert investigation workflow.
- Explain identity and access management (IAM) provisioning and the user lifecycle, including offboarding.
- Describe single sign-on (SSO), directory services, federation and identity governance.
- Compare multifactor authentication (MFA) factors, and explain privileged access management (PAM).
- Explain automation use cases and orchestration, with guardrails and human oversight.
- Describe the four incident response phases in order, and classify incidents and assign severity.
- Name the incident response roles and explain what each contributes.
- Compare tabletop, functional and full-scale exercises, and describe training and after-action review.
- Explain threat hunting methods: hypothesis-driven investigation and indicator sweeping.
- Apply root cause analysis in order, from problem identification to verifying the fix.
- Describe forensic collection and preservation: write blockers, forensic imaging and hash verification.
- Explain chain of custody, legal authorisation and privacy law (GDPR, ECPA).

- Identify key log sources, and explain collection, centralisation, correlation and retention.
- Describe network capture, memory forensics and metadata analysis as sources beyond logs.
- Recognise indicators of malicious activity: malware, network, application and behavioural.

CompTIA exam context

Item	Detail
Exam	Network Fundamentals + CompTIA Security+ SY0-701
Objective area	Domain 4 — Security Operations (28%, the largest domain)
Pass mark	750 on a scale of 100–900
Style	Scenario questions: which control at which layer, which IR phase, which evidence step
Practical	Performance-based items may test IR sequencing and evidence handling

Chapter coverage

Chapter	Title	What is covered
1	Monitoring and Network Security Devices	Firewall types; rule management and segmentation; WAF; IDS and IPS; detection methods; reading a host session log
2	Endpoint and Data Controls	DNS filtering; DLP; NAC; EDR and XDR; the alert investigation workflow
3	Identity, Access and Automation	Provisioning and the user lifecycle; SSO, directories and federation; MFA; PAM; automation and orchestration
4	Incident Response	The four phases; classification, severity and roles; exercises; training and after-action review
5	Threat Hunting and Forensic Evidence	Hunting methods; root cause analysis; collecting and preserving evidence; chain of custody and legal requirements
6	Logs, Investigation and Consolidation	Log sources; collection, centralisation and retention; captures, memory and metadata; takeaways

How this session maps to the exam

Every activity and quiz question is tagged with the exam objective it tests (for example 11.3 Forensics). Using the session quiz to 70%+ before the exam is the fastest route to Domain 4 marks.